

Course Outcomes (COs)	Course Outcome Statements
CO1	Explain the foundational concepts of Artificial Intelligence, including its inception, intelligent agents, and types of task environments.
CO2	Trace uninformed, informed, local, and adversarial search strategies for solving AI problems.
CO3	Perform manual logical inference using propositional and predicate logic by considering the methods of knowledge representation
CO4	Apply probabilistic reasoning and basic fuzzy-logic concepts to handle uncertainty in AI systems.
CO5	analyze the foundational machine learning paradigms underlying the design and implementation of AI-based systems.

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
CO1	<i>TSO 1a.</i> Explain the evolution and foundational philosophies of AI. <i>TSO 1b.</i> Describe the structure and functioning of intelligent agents using the PEAS framework. <i>TSO 1c.</i> Classify task environments based on their properties. <i>TSO 1d.</i> Differentiate between types of agents based on their internal architecture. <i>TSO 1e.</i> Outline AI application domains across engineering disciplines.	Unit- 1.0: Introduction to Artificial Intelligence (8hrs) 1.1 What is AI? History and evolution of AI 1.2 Foundations of AI: philosophy, mathematics, and cognitive-science influences 1.3 Intelligent agents: agent–environment interaction, PEAS (Performance measure, Environment, Actuators, and Sensors) description 1.4 Types of environments (observable, deterministic, episodic, static, discrete, single/multi-agent) 1.5 Types of agents: simple reflex, model-based, goal-based, utility-based, learning agents 1.6 Overview of AI application domains across engineering disciplines
CO2	<i>TSO 2a.</i> Represent a given real-world problem as a state-space search problem.	Unit- 2.0: Problem Solving by Search (9hrs)

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	<i>TSO 2b.</i> Trace uninformed search strategies manually and compare their complexity. <i>TSO 2c.</i> Trace informed search strategies using suitable heuristic functions. <i>TSO 2d.</i> Verify the admissibility and consistency of a given heuristic. <i>TSO 2e.</i> Illustrate local search techniques for simple optimisation problems. <i>TSO 2f.</i> Trace adversarial search with alpha-beta pruning for a two-player game. <i>TSO 2g.</i> Illustrate simple Constraint Satisfaction Problems (CSPs).	2.1 Problem formulation and state-space representation 2.2 Uninformed search: BFS, DFS, Uniform Cost Search, Depth-Limited and Iterative Deepening Search 2.3 Informed (heuristic) search: Greedy Best-First Search, A* algorithm 2.4 Heuristic function design and admissibility/consistency 2.5 Local search algorithms: Hill Climbing, Simulated Annealing (concept) 2.6 Adversarial search: Minimax algorithm, Alpha-Beta pruning 2.7 Constraint Satisfaction Problems (CSP): concept, backtracking (overview)
CO3	<i>TSO 3a.</i> Represent simple real-world facts using propositional and predicate logic. <i>TSO 3b.</i> Apply resolution manually to perform logical inference. <i>TSO 3c.</i> Trace forward and backward chaining to derive conclusions from a knowledge base. <i>TSO 3d.</i> Represent structured knowledge using semantic networks and frames. <i>TSO 3e.</i> Outline the design of a simple rule-based (production) system for a given domain.	Unit- 3.0: Knowledge Representation and Reasoning (8hrs) 3.1 Knowledge representation issues; propositional logic — syntax and semantics 3.2 Inference in propositional logic: resolution 3.3 First-order predicate logic — syntax, quantifiers (concept) 3.4 Inference in first-order logic: forward chaining, backward chaining, resolution (concept) 3.5 Structured knowledge representation: semantic networks, frames 3.6 Rule-based (production) systems
CO4	<i>TSO 4a.</i> Explain the need for probabilistic reasoning in AI. <i>TSO 4b.</i> Apply Bayes' theorem to compute conditional probabilities in AI problems. <i>TSO 4c.</i> Construct a simple Bayesian network for a given scenario. <i>TSO 4d.</i> Explain the basic principles of fuzzy logic and fuzzy-set operations.	Unit- 4.0: Reasoning Under Uncertainty (7hrs) 4.1 Sources of uncertainty; review of basic probability theory 4.2 Bayes' theorem and its application to AI reasoning 4.3 Bayesian networks: representation and manual inference (overview) 4.4 Introduction to fuzzy logic and fuzzy sets 4.5 Certainty factors and rule-based uncertainty (overview)

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
CO5, CO6	<i>TSO 5a.</i> Differentiate between supervised, unsupervised, and reinforcement learning paradigms. <i>TSO 5b.</i> Describe the working principle of basic regression, classification, and clustering techniques. <i>TSO 5c.</i> Describe the basic structure of a perceptron and an artificial neural network. <i>TSO 5d.</i> Describe applications of AI in natural language processing, computer vision, and expert systems. <i>TSO 5e.</i> Identify ethical, bias-related, and societal implications of AI deployment. <i>TSO 5f.</i> Relate AI applications to sustainable development goals and multidisciplinary engineering contexts.	Unit- 5.0: Introduction to Machine Learning and AI Applications (10hrs) 5.1 Machine learning paradigms: supervised, unsupervised, reinforcement learning (concept) 5.2 Overview of regression and classification concepts (linear regression, k-NN, decision trees) 5.3 Overview of clustering concept (k-means) 5.4 Introduction to artificial neural networks: perceptron, basic architecture (concept) 5.5 Overview of Natural Language Processing (NLP): concept and applications 5.6 Overview of Computer Vision: concept and applications 5.7 Expert systems: architecture and applications 5.8 Robotics and AI: overview of intelligent robotic systems 5.9 Ethics, bias, and responsible AI; societal impact; AI and Sustainable Development Goals (SDGs)

Reference/Books:

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
1.	Artificial Intelligence: A Modern Approach	Stuart Russell, Peter Norvig	Pearson; 4th Ed., 2020. ISBN: 978-0134610993
2.	AI for everyone	Saptarsi Goswami, Amit Kumar Das, Amlan Chakrabarti	Pearson Education: SBN-10 : 9361591754, ISBN-13: 978-9361591754
3.	Introduction to Artificial Intelligence and Expert Systems	Dan W. Patterson	PHI Learning; 2015. ISBN: 978-8120307775
4.	Artificial Intelligence	Elaine Rich, Kevin Knight, Shivashankar B. Nair	McGraw-Hill; 3rd Ed., 2017. ISBN: 978-1259029420

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
5.	Artificial Intelligence - A Modern Approach	Dilip Narayan Sahu, Kishore Kumar Dash, Abahy Kolhe, Agilandeswari Devarajan	ISBN-10: 9355151446 ISBN-13: 978-9355151445
6.	Machine Learning Essentials	Dhairya Parikh	Vibrant Publishers: ISBN-10: 1636513778, ISBN-13: 978-1636513775